

Summary

Mathematics graduate with strong analytical foundation in Python, SQL, and Machine Learning. Experienced in building end-to-end data science projects, including collaborative filtering recommendation systems, with hands-on skills in data analysis, statistical modeling, and BI visualization (Power BI, Tableau).

Education

Siddharth University, Kapilvastu, Siddharth Nagar <i>Master of Science (M.Sc.) in Mathematics</i>	2023 – 2025
Siddharth University, Kapilvastu, Siddharth Nagar <i>Bachelor of Science (B.Sc.)</i>	2020 – 2023

Technical Skills

Languages: Python, SQL
Libraries: Pandas, NumPy, Matplotlib, Seaborn
Tools: Excel, Power BI, Tableau
Concepts: Window Functions, Hypothesis Testing, Probability, Classification, Regression, Clustering

Projects

Retail Data Warehouse & ML Optimization Platform | *Python, SQL, XGBoost, Streamlit, scikit-learn*

- Architected an end-to-end Data Warehousing + BI + ML pipeline: built a star-schema SQLite warehouse (fact/dimension tables) processing 95,000+ retail transactions across 500 customers and 60 products.
- Built an XGBoost revenue forecasting model (under 20% MAPE) using lag, rolling-window, and calendar features, plus a log-log price-elasticity regression recovering category-level elasticities within 5% of ground truth.
- Developed a leakage-free customer churn classifier (XGBoost, temporal train/test split), RFM + K-Means customer segmentation, and IsolationForest/rolling z-score anomaly detection flagging revenue and transaction outliers.
- Engineered an interactive Streamlit BI dashboard with dynamic filtering, a live price-elasticity what-if simulator, one-click CSV ingestion that auto-rebuilds the warehouse and retrains all 5 ML models, and automated Excel/PDF report export.

Movie Recommendation System | *Python, NumPy, Streamlit, Scikit-Learn*

- Built a movie recommendation engine from scratch using low-rank Matrix Factorization (SGD-based Collaborative Filtering) on the MovieLens dataset (100,836 ratings, 610 users, 9,742 movies).
- Achieved 15.75% RMSE improvement (0.8788 vs. 1.0432 baseline) and MAE of 0.6724 via user/item bias modeling and residual diagnostics.
- Engineered end-to-end ML pipeline (data preprocessing, EDA, model serialization) modularized across 7 Python modules; validated with Pytest unit tests.
- Deployed an interactive Streamlit web app for real-time, personalized unseen-movie recommendations.

Sales Analytics Pipeline | *Python, SQL, Excel*

- Designed a normalized SQLite schema and wrote analytical SQL (window functions, CTEs, joins) for RFM customer segmentation and regional profit analysis across 8,000+ orders.
- Built a Python ETL pipeline (pandas) to load and transform sales data, then auto-generated a multi-sheet Excel dashboard (openpyxl) with embedded charts.
- Automated churn-risk identification by flagging customers inactive 180+ days using SQL date functions, replacing manual tracking with a repeatable query.